

MINIMUM VARIANCE UNBIASED ESTIMATION IN A MULTIVARIATE LINEAR REGRESSION

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Keywords:

Rao-Blackwell-Lehmann- Scheffe theorem.

Summary:

The problem of minimum variance unbiased estimation of a multivariate normal pdf in a multivariate linear regression model is considered. Using the Rao-Blackwell-Lehman-Scheffe theorem the minimum variance unbiased estimators of the pdf are obtained under different assumptions about the covariance matrix. It is shown that the minimum variance unbiased estimators obtained in Ghurye and Olkin (1969) are special cases of our estimators. We also correct the results of O'Reilly and Zarate (1984).

1. INTRODUCTION

In recent years many researchers have considered the problem of constructing minimum variance unbiased estimators (MVUEs) of an unknown probability density function (pdf) and its functionals. The earliest work in this area was that of Kolmogorov (1950) who, using the Rao-Blackwell-Lehmann-Scheffe (abbreviated RBLS) theorem, presented a general method for deriving the MVUE of an unknown pdf as a function of the complete sufficient statistic. Washio, Morimoto and Ikeda (1956), Barton (1961), Patil (1963) and Chhikara and Folks (1974), among others, used the RBLS theorem to derive the MVUEs of the pdfs corresponding to many probability models that include the Binomial, Poisson, Normal and the Inverse Gaussian distributions. Using the same theorem, Ghurye and Olkin (1969) and Lumel'skij and Sapozhnikov (1969) independently

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obtained the MVUEs of the multivariate normal probability density function under various assumptions about the mean vector and the covariance matrix. Ghurye and Olkin (1969) also considered the estimation of a multivariate normal pdf when the mean vector is linearly restricted to a lower dimensional subspace and their results include the MVUE of a univariate normal pdf in a linear regression model as a special case. O'Reilly and Zarate (1984) found the MVUE of the multivariate normal pdf in the case of multivariate normal regression. However, their results have minor errors.

In this paper we consider the problem of minimum variance unbiased estimation of a multivariate normal pdf in a multivariate linear regression model under various assumptions about the covariance matrix. It is found that the MVUEs of the multivariate normal pdf and the univariate normal pdf in a linear regression model obtained in Ghurye and Olkin (1969) are special cases of our estimators, and we also correct the results of O'Reilly and Zarate (1984).

2. NOTATION

All vectors are column vectors unless otherwise indicated, and are indicated in bold face. All matrices are indicated by bold face.

2.1 Model Parameters and Statistics

$\mathbf{Y}_\alpha = (Y_{\alpha 1}, \dots, Y_{\alpha p})$ = responses of α^{th} experimental unit (row vector)

$$\mathbf{Y} = \begin{bmatrix} Y_{11}, \dots, Y_{1p} \\ \vdots \\ Y_{n1}, \dots, Y_{np} \end{bmatrix} = \text{data matrix of observed responses}$$

$\mathbf{X}_\alpha = (X_{\alpha 1}, \dots, X_{\alpha q})$ = design vector for α^{th} experimental unit (row vector)

$$\mathbf{X} = \begin{bmatrix} X_{11}, \dots, X_{1q} \\ \vdots \\ X_{n1}, \dots, X_{nq} \end{bmatrix} = \text{design matrix}$$

$$\mathbf{B} = \begin{bmatrix} \beta_{11}, \dots, \beta_{1p} \\ \vdots \\ \beta_{q1}, \dots, \beta_{qp} \end{bmatrix} = \text{parameter matrix}$$

$$\mathbf{\Sigma} = \begin{bmatrix} \sigma_{11}, \dots, \sigma_{1p} \\ \vdots \\ \sigma_{p1}, \dots, \sigma_{pp} \end{bmatrix} = \text{variance-covariance matrix}$$

$\epsilon_\alpha = (\epsilon_{\alpha 1}, \dots, \epsilon_{\alpha p}) = \text{random component of } \alpha^{\text{th}} \text{ experimental unit (row vector)}$

$$\mathbf{E} = \begin{bmatrix} \epsilon_{11}, \dots, \epsilon_{1p} \\ \vdots \\ \epsilon_{n1}, \dots, \epsilon_{np} \end{bmatrix} = \text{matrix of random components.}$$

2.2 Matrix Manipulations

$\text{vec}(\mathbf{A})$ is the vectorization of the matrix $\mathbf{A}$ obtained by stacking the columns of $\mathbf{A}$.

$\mathbf{A} \otimes \mathbf{B}$ is the right Kronecker (or direct) product of matrices $\mathbf{A}$ and $\mathbf{B}$.

$|\mathbf{A}|$ determinant of $\mathbf{A}$

$\Psi(\mathbf{A}) = |\mathbf{A}| \text{ if } |\mathbf{A}| > 0$
0 otherwise

$\text{exptr}(\mathbf{A}) = \exp(\text{trace}(\mathbf{A}))$

$\mathbf{I}$ identity matrix

$\mathbf{J}$ matrix whose components are all ones

$\mathbf{1}$ a vector whose components are all ones.

2.3 Distributions of vector and matrix random variables

$MVN_r(\mu, \Sigma)$ is the r -dimensional multivariate normal distribution with mean vector μ and covariance-matrix Σ .
If $Z \sim MVN_r(\mu, \Sigma)$ then

$$\begin{aligned} f_Z(z) &= |2\pi\Sigma|^{-\frac{1}{2}} \exp \left\{ -\frac{1}{2} \text{vec}(z - \mu)' \Sigma^{-1} \text{vec}(z - \mu) \right\} \\ &= |2\pi\Sigma|^{-\frac{1}{2}} \exp \left\{ -\frac{1}{2} \text{trace} \left\{ \Sigma^{-1} \text{vec}(z - \mu) \text{vec}(z - \mu)' \right\} \right\} \end{aligned}$$

$W(p, \nu, \Sigma)$ is the Wishart distribution with ν degrees of freedom and $p \times p$ matrix Σ

2.4 Other notation

$d(p, \nu)$ is the function

$$d(p, \nu) = 2^{p\nu/2} \pi^{p(p-1)/4} \prod_{i=1}^p \Gamma \left(\frac{1}{2}(\nu - i + 1) \right)^{-1}$$

3. Σ IS COMPLETELY KNOWN

The model to be considered is

$$Y = XB + E$$

where $\epsilon_i \sim MVN_p(0, \Sigma)$, ϵ_i is independent of ϵ_j ($i \neq j$) and Σ is known completely. By examining the likelihood (Appendix B) we see that $X'Y$ is a sufficient statistic for B . It is well known that the distribution of $\text{vec}(X'Y)$ is

$$\text{vec}(X'Y) \sim MVN_{qp}(\text{vec}(X'XB), \Sigma \otimes X'X).$$

Hence, $X'Y$ is a complete sufficient statistic for B when Σ is known, and by the Rao-Blackwell-Lehmann-Scheffe theorem, the MVUE of the density for Y_α at a given X_α is found as the conditional pdf of Y_α given $X'Y$. In Appendix C this is found to be:

$$\begin{aligned} & (2\pi)^{-p/2} |\Sigma|^{-\frac{1}{2}} |I - X_\alpha(X'X)^{-1}X'_\alpha|^{-p/2} \times \\ & \exp \left\{ -\frac{1}{2} \Sigma^{-1} \left[(Y_\alpha - X_\alpha \hat{B})' (I - X_\alpha(X'X)^{-1}X'_\alpha)^{-1} (Y_\alpha - X_\alpha \hat{B}) \right] \right\} \end{aligned}$$

which agrees with O'Reilly and Zarate (1984) after a minor correction to their results (their results are missing the inverse of the middle term in the $\exp$ trace component).

When $q = 1$, and $\mathbf{X} = \mathbf{1}_{n \times 1}$, then $\mathbf{Y}_1, \dots, \mathbf{Y}_n$ are independent and identically distributed p -dimensional normal random variables with mean $\mathbf{B}_{1 \times p}$ and covariance matrix Σ . In this case, the above estimator reduces to

$$(2\pi)^{-p/2} |\Sigma|^{-1} ((n-1)/n)^{-p/2} \exp \left\{ -\frac{1}{2} \frac{n}{(n-1)} \Sigma^{-1} (\mathbf{Y}_\alpha - \bar{\mathbf{Y}})' (\mathbf{Y}_\alpha - \bar{\mathbf{Y}}) \right\}$$

where $\bar{\mathbf{Y}} = \frac{1}{n} \sum_{i=1}^n \mathbf{Y}_i$ which agrees with Result 3.1 of Ghurye and Olkin (1969) and Result 1 of Lumel'skij and Sapozhnikov (1969).

4. $\Sigma = \sigma^2 \Sigma_0$ WHERE Σ_0 IS KNOWN AND σ^2 IS UNKNOWN

The model to be considered is now:

$$\mathbf{Y} = \mathbf{XB} + \mathbf{E}$$

where $\epsilon_i \sim \text{MVN}_p(0, \sigma^2 \Sigma_0)$; ϵ_i is independent of $\epsilon_j (i \neq j)$, Σ_0 is known completely, and σ^2 is unknown. From the likelihood in Appendix B, we see that $\mathbf{X}'\mathbf{Y}$ and $\text{trace}(\Sigma_0^{-1} \hat{\Sigma})$ are sufficient statistics for $\mathbf{B}$ and σ^2 and are independent. Now $\text{trace}(n\Sigma_0^{-1} \hat{\Sigma}) \sim W(1, p(n-q), \sigma^2)$, and hence is complete.

According to lemma 1 (Appendix A) we first find the MVUE of the pdf of $\mathbf{Y}_\alpha$ for a fixed $\mathbf{X}_\alpha$ when σ^2 is known (Section 3), and then find a MVUE of the function of σ^2 by using Lemma 2 (Appendix A). By simple substitution, the MVUE of the density function for $\mathbf{Y}_\alpha$ for a fixed $\mathbf{X}_\alpha$ is found to be:

$$\begin{aligned} & (\pi)^{-p/2} |\mathbf{I} - \mathbf{X}_\alpha (\mathbf{X}'\mathbf{X})^{-1} \mathbf{X}'_\alpha|^{-p/2} \frac{\Gamma(\frac{1}{2}p(n-q))}{\Gamma(\frac{1}{2}p(n-q-1))} \\ & \left[\text{trace}(n\Sigma_0^{-1} \hat{\Sigma}) \right]^{-(p(n-q)-2)/2} \times \\ & \Psi \left[\text{trace} \Sigma_0^{-1} \{ n\hat{\Sigma} - (\mathbf{Y}_\alpha - \mathbf{X}_\alpha \hat{\mathbf{B}})' (\mathbf{I} - \mathbf{X}_\alpha (\mathbf{X}'\mathbf{X})^{-1} \mathbf{X}'_\alpha)^{-1} \right. \\ & \left. (\mathbf{Y}_\alpha - \mathbf{X}_\alpha \hat{\mathbf{B}}) \} \right]^{(p(n-q-1)-2)/2} \end{aligned}$$

When $q = 1$ and $\mathbf{X} = \mathbf{1}_{n \times 1}$, the above estimator reduces to

$$\frac{(n)^{p/2}}{(n-1)^{p/2} \pi^{p/2} |\Sigma_0|^{\frac{1}{2}} \Gamma\left(\frac{1}{2}p(n-1)\right)} \times \\ \left[\text{trace} \Sigma_0^{-1} (\mathbf{Y} - \mathbf{1}\bar{\mathbf{Y}})' (\mathbf{Y} - \mathbf{1}\bar{\mathbf{Y}}) \right]^{-(p(n-1)-2)/2} \times \\ \Psi \left[\text{trace} \Sigma_0^{-1} \left\{ (\mathbf{Y} - \mathbf{1}\bar{\mathbf{Y}})' (\mathbf{Y} - \mathbf{1}\bar{\mathbf{Y}}) - \right. \right. \\ \left. \left. \frac{n}{n-1} (\mathbf{Y}_\alpha - \bar{\mathbf{Y}})' (\mathbf{Y}_\alpha - \bar{\mathbf{Y}}) \right\} \right]^{(p(n-2)-2)/2}$$

which agrees with Result 3.2 of Ghurye and Olkin (1969).

When $p = 1$, we are in the case of univariate multiple regression and the estimator reduces to:

$$(\pi)^{-\frac{1}{2}} |\mathbf{I} - \mathbf{X}_\alpha (\mathbf{X}'\mathbf{X})^{-1} \mathbf{X}'_\alpha|^{-\frac{1}{2}} \frac{\Gamma\left(\frac{1}{2}(n-q)\right)}{\Gamma\left(\frac{1}{2}p(n-q-1)\right)} (n\hat{\sigma}^2)^{-(n-q-2)/2} \times \\ \Psi \left[n\hat{\sigma}^2 - (\mathbf{Y}_\alpha - \mathbf{X}_\alpha \hat{\mathbf{B}})' (\mathbf{I} - \mathbf{X}_\alpha (\mathbf{X}'\mathbf{X})^{-1} \mathbf{X}'_\alpha)^{-1} (\mathbf{Y}_\alpha - \mathbf{X}_\alpha \hat{\mathbf{B}}) \right]^{(n-q-3)/2}$$

which (after various simplifications) agrees with result 4.2 of Ghurye and Olkin (1969).

5. INTRA-CLASS CORRELATION MODEL $\Sigma = \sigma^2(1-\rho)\mathbf{I} + \sigma^2\rho\mathbf{J}$

The model to be considered is now:

$$\mathbf{Y} = \mathbf{XB} + \mathbf{E}$$

where $\epsilon_i \sim \text{MVN}_p(0, \sigma^2(1-\rho)\mathbf{I} + \sigma^2\rho\mathbf{J})$, ϵ_i is independent of ϵ_j ($i \neq j$), and σ^2 and ρ are unknown ($-(p-1)^{-1} < \rho < 1$). From the likelihood in Appendix B, we see that $\mathbf{X}'\mathbf{Y}$, $U = n \cdot \text{trace}(\hat{\Sigma}\mathbf{J}/p)$ and $V = n \cdot \text{trace}(\hat{\Sigma}(\mathbf{I}-\mathbf{J}/p))$ are sufficient statistics for $\mathbf{B}$, $\delta^{-1} = \sigma^2(1+(p-1)\rho)$, and $\eta^{-1} = \sigma^2(1-\rho)$, and are independent. Furthermore, $U \sim W(1, n-q, \delta^{-1})$ and $V \sim W(1, (p-1)(n-q), \eta^{-1})$. Hence the sufficient statistics are complete.

According to lemma 1, we first find a MVUE of the pdf assuming σ^2 and ρ are known (Section 3). We then apply lemma 2 twice to find the MVUE of the two functions of σ^2 and ρ . The MVUE of the density function for $\mathbf{Y}_\alpha$ for a fixed $\mathbf{X}_\alpha$ is found to be:

$$\begin{aligned} & (\pi)^{-p/2} |\mathbf{I} - \mathbf{X}_\alpha (\mathbf{X}'\mathbf{X})^{-1} \mathbf{X}'_\alpha|^{-p/2} \times \\ & \frac{\Gamma(\frac{1}{2}p(n-q))}{\Gamma(\frac{1}{2}p(n-q-1))} \frac{\Gamma(\frac{1}{2}(p-1)(n-q))}{\Gamma(\frac{1}{2}(p-1)(n-q-1))} \times \\ & \text{trace}(n\hat{\Sigma}\mathbf{J}/p)^{-(n-q-2)/2} \text{trace}(n\hat{\Sigma}(\mathbf{I} - \mathbf{J}/p))^{-((p-1)(n-q)-2)/2} \times \\ & \Psi \left[\text{trace} \left[(n\hat{\Sigma} - (\mathbf{Y}_\alpha - \mathbf{X}_\alpha \hat{\mathbf{B}})'(\mathbf{I} - \mathbf{X}_\alpha (\mathbf{X}'\mathbf{X})^{-1} \mathbf{X}'_\alpha)^{-1} \times \right. \right. \\ & \left. \left. (\mathbf{Y}_\alpha - \mathbf{X}_\alpha \hat{\mathbf{B}}))\mathbf{J}/p \right] \right]^{(n-q-3)/2} \times \\ & \Psi \left[\text{trace} \left[(n\hat{\Sigma} - (\mathbf{Y}_\alpha - \mathbf{X}_\alpha \hat{\mathbf{B}})'(\mathbf{I} - \mathbf{X}_\alpha (\mathbf{X}'\mathbf{X})^{-1} \mathbf{X}'_\alpha)^{-1} \times \right. \right. \\ & \left. \left. (\mathbf{Y}_\alpha - \mathbf{X}_\alpha \hat{\mathbf{B}}))(\mathbf{I} - \mathbf{J}/p) \right] \right]^{((p-1)(n-q)-2)/2}. \end{aligned}$$

When $q = 1$ and $\mathbf{X} = \mathbf{1}_{n \times 1}$, the above estimator reduces to

$$\begin{aligned} & \frac{(n)^{p/2}}{(n-1)^{p/2} \pi^{p/2}} \frac{\Gamma(\frac{1}{2}(n-1))}{\Gamma(\frac{1}{2}(n-2))} \frac{\Gamma(\frac{1}{2}(p-1)(n-1))}{\Gamma(\frac{1}{2}(p-1)(n-2))} \times \\ & \text{trace}[(\mathbf{Y} - \mathbf{1}\bar{\mathbf{Y}})'(\mathbf{Y} - \mathbf{1}\bar{\mathbf{Y}})\mathbf{J}/p]^{-(n-3)/2} \times \\ & \text{trace}[(\mathbf{Y} - \mathbf{1}\bar{\mathbf{Y}})'(\mathbf{Y} - \mathbf{1}\bar{\mathbf{Y}})(\mathbf{I} - \mathbf{J}/p)]^{-((p-1)(n-1)-2)/2} \times \\ & \Psi \left[\text{trace} \left\{ \{(\mathbf{Y} - \mathbf{1}\bar{\mathbf{Y}})'(\mathbf{Y} - \mathbf{1}\bar{\mathbf{Y}}) - \frac{n}{n-1}(\mathbf{Y}_\alpha - \bar{\mathbf{Y}})'(\mathbf{Y}_\alpha - \bar{\mathbf{Y}})\mathbf{J}/p \} \right\} \right]^{(n-4)/2} \times \\ & \Psi \left[\text{trace} \left\{ \{(\mathbf{Y} - \mathbf{1}\bar{\mathbf{Y}})'(\mathbf{Y} - \mathbf{1}\bar{\mathbf{Y}}) - \frac{n}{n-1}(\mathbf{Y}_\alpha - \bar{\mathbf{Y}})' \times \right. \right. \\ & \left. \left. (\mathbf{Y}_\alpha - \bar{\mathbf{Y}})\mathbf{J}/p \} \right\} \right]^{((p-1)(n-2)-2)/2} \end{aligned}$$

which agrees with Result 3.3 of Ghurye and Olkin (1969).

6. Σ IS COMPLETELY UNKNOWN

The model to be considered is now:

$$\mathbf{Y} = \mathbf{X}\mathbf{B} + \mathbf{E}$$

where $\epsilon_i \sim \text{MVN}_p(0, \Sigma)$, ϵ_i is independent of ϵ_j ($i \neq j$) and Σ is completely unknown. By examining the likelihood, we see that $\hat{\mathbf{B}}$ and $\hat{\Sigma}$ are sufficient for $\mathbf{B}$ and Σ and are independent. Also, $n\hat{\Sigma} \sim W(p, n - q, \Sigma)$ and hence is complete. We again find the MVUE of the pdf when Σ is known (Section 3), and then use lemma 2 to find the MVUE of this function of Σ . The estimate of the density function for $\mathbf{Y}_\alpha$ for a fixed $\mathbf{X}_\alpha$ is then:

$$(\pi)^{-p/2} |\mathbf{I} - \mathbf{X}_\alpha (\mathbf{X}'\mathbf{X})^{-1} \mathbf{X}'_\alpha|^{-p/2} \frac{d(p, n - q - 1)}{d(p, n - q)} |n\hat{\Sigma}|^{-(n-p-q-1)/2} \times \\ \Psi \left[n\hat{\Sigma} - (\mathbf{Y}_\alpha - \mathbf{X}_\alpha \hat{\mathbf{B}})' (\mathbf{I} - \mathbf{X}_\alpha (\mathbf{X}'\mathbf{X})^{-1} \mathbf{X}'_\alpha)^{-1} (\mathbf{Y}_\alpha - \mathbf{X}_\alpha \hat{\mathbf{B}}) \right]^{(n-p-q-2)/2}$$

which agrees with Theorem 3.1 of O'Reilly and Zarate (1984) after a minor correction (they have an incorrect exponent of their determinants).

When $q = 1$, $\mathbf{X} = \mathbf{1}_{n \times 1}$, the above reduces to

$$(\pi)^{-p/2} ((n-1)/n)^{-p/2} \frac{\Gamma(\frac{1}{2}(n-1))}{\Gamma(\frac{1}{2}(n-p-1))} [(\mathbf{Y} - \mathbf{1}\bar{\mathbf{Y}})'(\mathbf{Y} - \mathbf{1}\bar{\mathbf{Y}})]^{-(n-p-2)/2} \times \\ \Psi \left[(\mathbf{Y} - \mathbf{1}\hat{\mathbf{Y}})'(\mathbf{Y} - \mathbf{1}\bar{\mathbf{Y}}) - \frac{n}{n-1} (\mathbf{Y}_\alpha - \bar{\mathbf{Y}})'(\mathbf{Y}_\alpha - \bar{\mathbf{Y}}) \right]^{(n-p-3)/2}$$

which agrees with Result 3.4 of Ghurye and Olkin (1969) and Result 1c of Lumel'skij and Sapozhnikov (1969).

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APPENDIX A

We use the following lemmas whose proofs are given in Ghurye and Olkin (1969).

Lemma 1:

Let S and T independent statistics with a joint cdf $F(t; \sigma, \tau)G(s; \sigma)$ depending on the parameters $(\sigma, \tau) \in \Omega_1 \times \Omega_2$. Let $b(\sigma, T)$ be an unbiased estimator of $\{h(\sigma, \tau), \tau \in \Omega_2\}$ for each $\sigma \in \Omega_1$ and $a(S, \tau)$ be an unbiased estimator of $\{b(\sigma, \tau), \sigma \in \Omega_1\}$ for each $\tau \in \Omega_2$. Then $a(S, T)$ is an unbiased estimator of $\{h(\sigma, \tau), (\sigma, \tau) \in \Omega_1 \times \Omega_2\}$.

Lemma 2:

For any $\mathbf{G} \geq 0$ and any real $m < \nu - p + 1$, if $\mathbf{S} \sim W(p, \nu, \Sigma)$, then the MVUE of $|\Sigma|^{-m/2} \exp \left[-\frac{1}{2} \text{trace} \Sigma^{-1} \mathbf{G} \right]$ is

$$\frac{d(p, \nu - m)}{d(p, \nu)} |\mathbf{S}|^{-(\nu - p - 1)/2} [\Psi(\mathbf{S} - \mathbf{G})]^{(\nu - m - p - 1)/2}.$$

APPENDIX B : LIKELIHOOD OF MULTIVARIATE REGRESSION

The model for a multivariate regression is:

$$\mathbf{Y} = \mathbf{XB} + \mathbf{E}$$

where ϵ_i are $MVN_p(0, \Sigma)$ and ϵ_i and ϵ_j ($i \neq j$) are independent. The log-likelihood is:

$$\log L(\mathbf{B}, \Sigma) = -\frac{1}{2} n \log |2\pi \Sigma| - \frac{1}{2} \text{trace} \{ (\mathbf{y} - \mathbf{XB}) \Sigma^{-1} (\mathbf{y} - \mathbf{XB})' \}.$$

$$\text{Let } \hat{\mathbf{B}} = (\mathbf{X}'\mathbf{X})^{-1} \mathbf{X}'\mathbf{Y}$$

$$\hat{\Sigma} = n^{-1} \mathbf{Y}'(\mathbf{I} - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1} \mathbf{X}') \mathbf{Y}.$$

Then the log-likelihood can be written as:

$$\log L(\mathbf{B}, \Sigma) = -\frac{1}{2} n \log |2\pi \Sigma| - \frac{1}{2} n \text{trace} \Sigma^{-1} \hat{\Sigma} - \frac{1}{2} \text{trace} \Sigma^{-1} (\hat{\mathbf{B}} - \mathbf{B})' \mathbf{X}' \mathbf{X} (\hat{\mathbf{B}} - \mathbf{B}).$$

APPENDIX C: DERIVATION OF CONDITIONAL PDF OF SECTION 3

To find the MVUE of the density, it is necessary to find the conditional pdf of $\mathbf{Y}_\alpha$ given $\mathbf{X}'\mathbf{Y}$. Without loss of generality, consider the conditional pdf of $\mathbf{Y}_1$ conditional upon $\mathbf{X}'\mathbf{Y}$. We partition the matrices $\mathbf{X}$ and $\mathbf{Y}$ as follows:

$$\mathbf{X} = \begin{bmatrix} \mathbf{X}_1 \\ \cdots \\ \mathbf{X}_2 \end{bmatrix} \text{ where } \mathbf{X}_1 \text{ is a } 1 \times q \text{ row vector, } \mathbf{X}_2 \text{ is a } (n-1) \times q \text{ matrix}$$

$$\mathbf{Y} = \begin{bmatrix} \mathbf{Y}_1 \\ \cdots \\ \mathbf{Y}_2 \end{bmatrix} \text{ where } \mathbf{Y}_1 \text{ is a } 1 \times q \text{ row vector, } \mathbf{Y}_2 \text{ is a } (n-1) \times p \text{ matrix.}$$

Now

$$\mathbf{X}'\mathbf{Y} = \mathbf{X}'_1\mathbf{Y}_1 + \mathbf{X}'_2\mathbf{Y}_2.$$

We define

$$\mathbf{A} = \mathbf{X}'\mathbf{X}$$

$$\mathbf{C} = \mathbf{X}'\mathbf{Y}$$

$$\mathbf{F} = \mathbf{X}'_1\mathbf{Y}_1.$$

Since $\mathbf{Y}_1$ and $\mathbf{Y}_2$ are independent, $\mathbf{Y}_1$ and $\mathbf{X}'_2\mathbf{Y}_2$ are also independent, and the joint pdf of $\mathbf{Y}_1$ and $\mathbf{X}'_2\mathbf{Y}_2$ is

$$\begin{aligned} f_{\mathbf{Y}_1, \mathbf{X}'_2\mathbf{Y}_2}(\mathbf{y}_1, \mathbf{X}'_2\mathbf{y}_2) = & \\ & |2\pi\boldsymbol{\Sigma}^{-\frac{1}{2}} \exp\{\text{tr}\{-\frac{1}{2}\boldsymbol{\Sigma}^{-1}\text{vec}(\mathbf{y}_1 - \mathbf{X}_1\mathbf{B})\text{vec}(\mathbf{y}_1 - \mathbf{X}_1\mathbf{B})'\} \times \\ & |2\pi\boldsymbol{\Sigma} \otimes \mathbf{X}'_2\mathbf{X}_2)|^{-\frac{1}{2}} \times \\ & \exp\{\text{tr}\{-\frac{1}{2}(\boldsymbol{\Sigma} \otimes \mathbf{X}'_2\mathbf{X}_2)^{-1}\text{vec}(\mathbf{X}'_2\mathbf{y}_2 - \mathbf{X}'_2\mathbf{X}_2\mathbf{B})\text{vec}(\mathbf{X}'_2\mathbf{y}_2 - \mathbf{X}'_2\mathbf{X}_2\mathbf{B})'\}\}. \end{aligned}$$

Now consider the transformation

$$\mathbf{C} = \mathbf{X}'\mathbf{Y} = \mathbf{X}'_1\mathbf{Y}_1 + \mathbf{X}'_2\mathbf{Y}_2 = \mathbf{F} + \mathbf{X}'_2\mathbf{Y}_2.$$

The inverse transformation is

$$\mathbf{X}'_2 \mathbf{Y}_2 = \mathbf{X}' \mathbf{Y} - \mathbf{X}'_1 \mathbf{Y}_1 = \mathbf{C} - \mathbf{F}$$

and the Jacobian has the value of 1. Hence the joint pdf of $\mathbf{Y}_1$ and $\mathbf{C}$ is

$$\begin{aligned} f_{\mathbf{Y}_1, \mathbf{C}}(\mathbf{y}_1, \mathbf{c}) = & |2\pi\boldsymbol{\Sigma}|^{-\frac{1}{2}} \exp\left\{-\frac{1}{2}\boldsymbol{\Sigma}^{-1} \text{vec}(\mathbf{y}_1 - \mathbf{X}_1\mathbf{B}) \text{vec}(\mathbf{y}_1 - \mathbf{X}_1\mathbf{B})'\right\} \times \\ & |2\pi\boldsymbol{\Sigma} \otimes \mathbf{X}'_2\mathbf{X}_2|^{-\frac{1}{2}} \exp\left\{-\frac{1}{2}(\boldsymbol{\Sigma} \otimes \mathbf{X}'_2\mathbf{X}_2)^{-1} \text{vec}(\mathbf{c} - \mathbf{f} - \mathbf{X}'_2\mathbf{X}_2\mathbf{B}) \right. \\ & \left. \text{vec}(\mathbf{c} - \mathbf{f} - \mathbf{X}'_2\mathbf{X}_2\mathbf{B})'\right\} \end{aligned}$$

and the conditional pdf of $\mathbf{Y}_1$ given $\mathbf{C} = \mathbf{X}'\mathbf{Y}$ is then

$$\begin{aligned} f_{\mathbf{Y}_1|\mathbf{C}}(\mathbf{y}_1) &= f_{\mathbf{Y}_1, \mathbf{C}}(\mathbf{y}_1, \mathbf{c})/f_{\mathbf{C}}(\mathbf{c}) \\ &= |2\pi\boldsymbol{\Sigma}|^{-\frac{1}{2}} \exp\left\{-\frac{1}{2}\boldsymbol{\Sigma}^{-1} \text{vec}(\mathbf{y}_1 - \mathbf{X}_1\mathbf{B}) \text{vec}(\mathbf{y}_1 - \mathbf{X}_1\mathbf{B})'\right\} \times \\ & |2\pi\boldsymbol{\Sigma} \otimes \mathbf{X}'_2\mathbf{X}_2|^{-\frac{1}{2}} \exp\left\{-\frac{1}{2}(\boldsymbol{\Sigma} \otimes \mathbf{X}'_2\mathbf{X}_2)^{-1} \times \right. \\ & \left. \text{vec}(\mathbf{c} - \mathbf{f} - \mathbf{X}'_2\mathbf{X}_2\mathbf{B}) \text{vec}(\mathbf{c} - \mathbf{f} - \mathbf{X}'_2\mathbf{X}_2\mathbf{B})'\right\} \div \\ & |2\pi\boldsymbol{\Sigma} \otimes \mathbf{A}|^{-\frac{1}{2}} \exp\left\{-\frac{1}{2}(\boldsymbol{\Sigma} \otimes \mathbf{A})^{-1} \text{vec}(\mathbf{c} - \mathbf{AB}) \text{vec}(\mathbf{c} - \mathbf{AB})'\right\} \end{aligned}$$

The constant term is simplified as follows:

$$\begin{aligned} \frac{|2\pi\boldsymbol{\Sigma}|^{-\frac{1}{2}} |2\pi\boldsymbol{\Sigma} \otimes \mathbf{X}'_2\mathbf{X}_2|^{-\frac{1}{2}}}{|2\pi\boldsymbol{\Sigma} \otimes \mathbf{A}|^{-\frac{1}{2}}} &= \frac{(2\pi)^{-p/2} (2\pi)^{-pq/2} |\boldsymbol{\Sigma}|^{-\frac{1}{2}} |\boldsymbol{\Sigma} \otimes \mathbf{X}'_2\mathbf{X}_2|^{-\frac{1}{2}}}{(2\pi)^{-pq/2} |\boldsymbol{\Sigma} \otimes \mathbf{A}|^{-\frac{1}{2}}} \\ &= (2\pi)^{-p/2} \frac{|\boldsymbol{\Sigma}|^{-\frac{1}{2}} |\boldsymbol{\Sigma}|^{-q/2} |\mathbf{X}'_2\mathbf{X}_2|^{-p/2}}{|\boldsymbol{\Sigma}|^{-q/2} |\mathbf{A}|^{-p/2}} \\ &= (2\pi)^{-p/2} \frac{|\mathbf{A}|^{p/2}}{|\boldsymbol{\Sigma}|^{\frac{1}{2}} |\mathbf{X}'_2\mathbf{X}_2|^{p/2}} = (2\pi)^{-p/2} |\boldsymbol{\Sigma}|^{-\frac{1}{2}} |\mathbf{I} - \mathbf{X}_1\mathbf{A}^{-1}\mathbf{X}'_1|^{-p/2} \end{aligned}$$

The exponential term may be simplified by noting that $\text{trace}(\mathbf{Axx}') = \mathbf{x}'\mathbf{Ax}$ for any matrix $\mathbf{A}$ and column vector $\mathbf{x}$ giving:

$$\begin{aligned} \exp - \frac{1}{2} [& \text{vec}(\mathbf{y}_1 - \mathbf{X}_1\mathbf{B})' \boldsymbol{\Sigma}^{-1} \text{vec}(\mathbf{y}_1 - \mathbf{X}_1\mathbf{B}) + \\ & \text{vec}(\mathbf{c} - \mathbf{f} - \mathbf{X}'_2\mathbf{X}_2\mathbf{B})' (\boldsymbol{\Sigma} \otimes \mathbf{X}'_2\mathbf{X}_2)^{-1} \text{vec}(\mathbf{c} - \mathbf{f} - \mathbf{X}'_2\mathbf{X}_2\mathbf{B}) - \\ & \text{vec}(\mathbf{c} - \mathbf{AB})' (\boldsymbol{\Sigma} \otimes \mathbf{A})^{-1} \text{vec}(\mathbf{c} - \mathbf{AB})] \end{aligned}$$

Since $(\mathbf{B}' \otimes \mathbf{A})\text{vec}(\mathbf{X}) = \text{vec}(\mathbf{AXB})$ for conformable $\mathbf{A}, \mathbf{X}, \mathbf{B}$, and $(\mathbf{A} \otimes \mathbf{B})^{-1} = \mathbf{A}^{-1} \otimes \mathbf{B}^{-1}$ when $\mathbf{A}$ and $\mathbf{B}$ are invertible, the expression reduces to:

$$\begin{aligned} \text{exp} - \frac{1}{2} [& \text{vec}(\mathbf{y}_1 - \mathbf{X}_1\mathbf{B})'\Sigma^{-1}\text{vec}(\mathbf{y}_1 - \mathbf{X}_1\mathbf{B}) + \\ & \text{vec}(\mathbf{c} - \mathbf{f} - \mathbf{X}'_2\mathbf{X}_2\mathbf{B})'\text{vec}((\mathbf{X}'_2\mathbf{X}_2)^{-1}(\mathbf{c} - \mathbf{f} - \mathbf{X}'_2\mathbf{X}_2\mathbf{B})\Sigma^{-1}) - \\ & \text{vec}(\mathbf{c} - \mathbf{AB})'\text{vec}(\mathbf{A}^{-1}(\mathbf{c} - \mathbf{AB})\Sigma^{-1})] \end{aligned}$$

and since $\text{vec}(\mathbf{A}')'\text{vec}(\mathbf{B}) = \text{trace}(\mathbf{AB})$, the expression can be further reduced as:

$$\begin{aligned} \text{exptr} - \frac{1}{2} [& \text{vec}(\mathbf{y}_1 - \mathbf{X}_1\mathbf{B})\text{vec}(\mathbf{y}_1 - \mathbf{X}_1\mathbf{B})'\Sigma^{-1} + \\ & (\mathbf{c}' - \mathbf{f}' - \mathbf{B}'\mathbf{X}'_2\mathbf{X}_2)(\mathbf{X}'_2\mathbf{X}_2)^{-1}(\mathbf{c} - \mathbf{f} - \mathbf{X}'_2\mathbf{X}_2\mathbf{B})\Sigma^{-1} - \\ & (\mathbf{c}' - \mathbf{B}'\mathbf{A}')\mathbf{A}^{-1}(\mathbf{c} - \mathbf{AB})\Sigma^{-1}] . \end{aligned}$$

Finally, since $\mathbf{y}_1$ is a row vector $\text{vec}(\mathbf{y}_1)' = \mathbf{y}_1$ and recalling that $\mathbf{A} = \mathbf{X}'_1\mathbf{X}_1 + \mathbf{X}'_2\mathbf{X}_2$ gives:

$$\text{exptr} - \frac{1}{2} [\mathbf{y}'_1\mathbf{y}_1 + (\mathbf{c} - \mathbf{f})'(\mathbf{X}'_2\mathbf{X}_2)^{-1}(\mathbf{c} - \mathbf{f}) - \mathbf{c}'\mathbf{A}^{-1}\mathbf{c}] \Sigma^{-1}$$

which after further algebraic manipulation yields:

$$\text{exptr} - \frac{1}{2} \left[\Sigma^{-1}(\mathbf{y}_\alpha - \mathbf{X}_\alpha\hat{\mathbf{B}})'(\mathbf{I} - \mathbf{X}_\alpha(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'_\alpha)^{-1}(\mathbf{y}_\alpha - \mathbf{X}_\alpha\hat{\mathbf{B}}) \right] .$$